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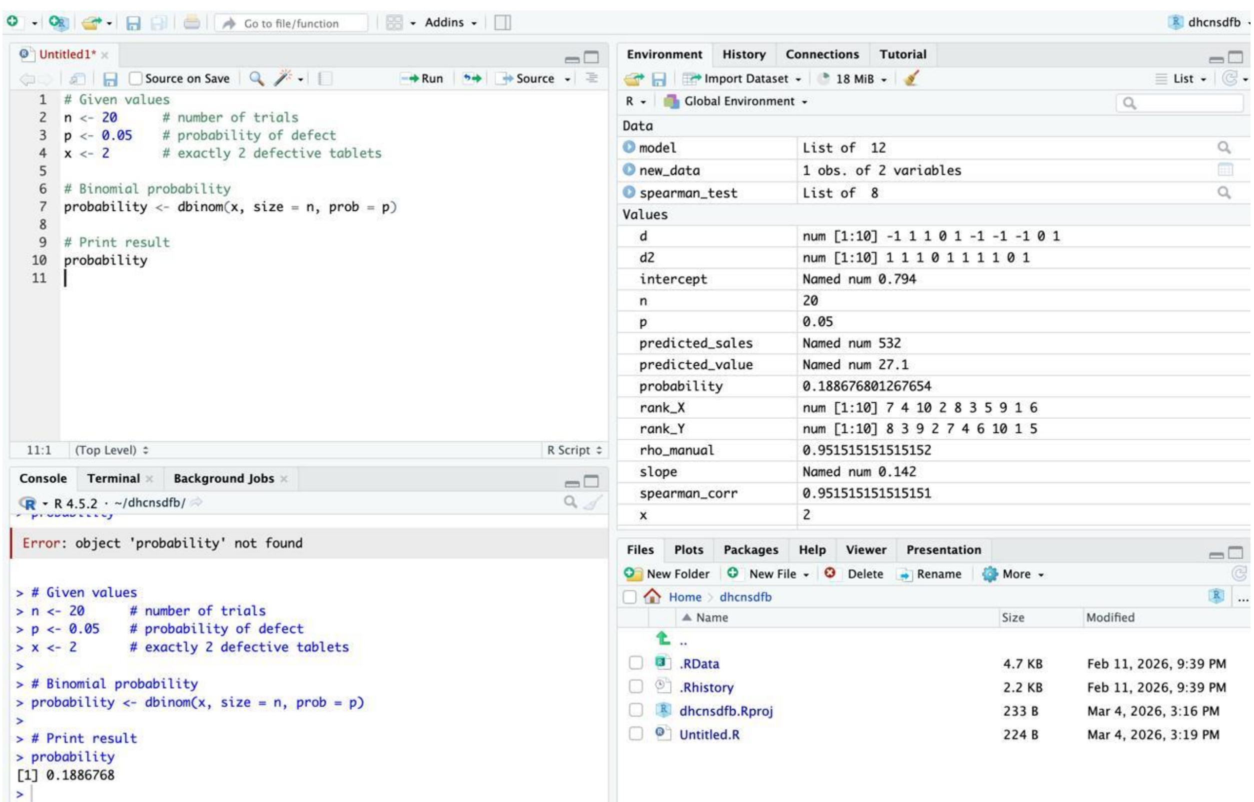
Probability and Statistics Lab (DA-3)

1. A pharmaceutical company reports that 5% of tablets produced are defective. A batch of 20 tablets is selected randomly for inspection. Solve using Binomial distribution to find the probability that exactly 2 tablets are defective.

CODE:



Que 1 (code part):-

Given values

n <- 20 # number of trials p <- 0.05 # probability of defect x <- 2 # exactly 2 defective tablets

Binomial probability

probability <- dbinom(x, size = n, prob = p)

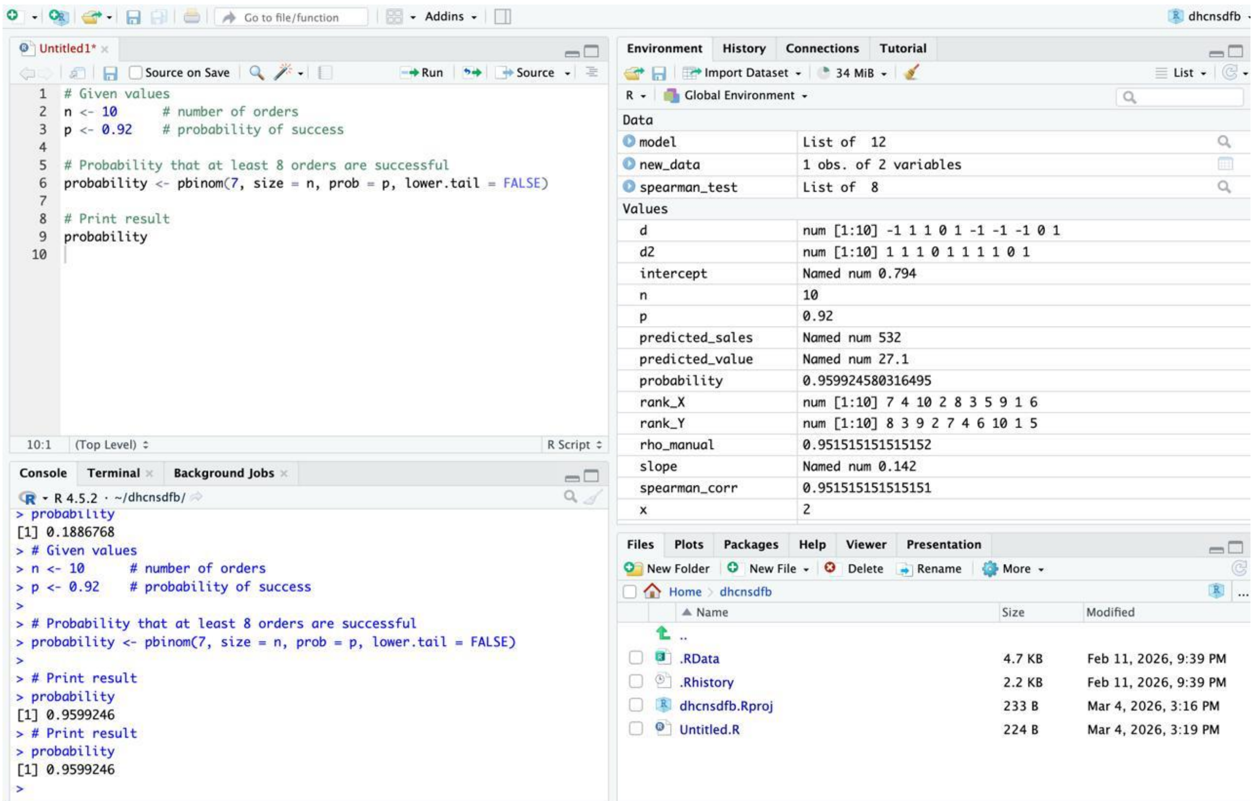
Print result

probability

QUE-2

1. In an online order processing system, the probability that an order is successfully completed without error is 0.92. If 10 orders are processed independently, solve using Binomial distribution to determine the probability that at least 8 orders are completed successfully.

CODE:-



Que –2(code part):-

Given values

n <- 10 # number of orders p <- 0.92 # probability of success

Probability that at least 8 orders are successful

probability <- pbinom(7, size = n, prob = p, lower.tail = FALSE)

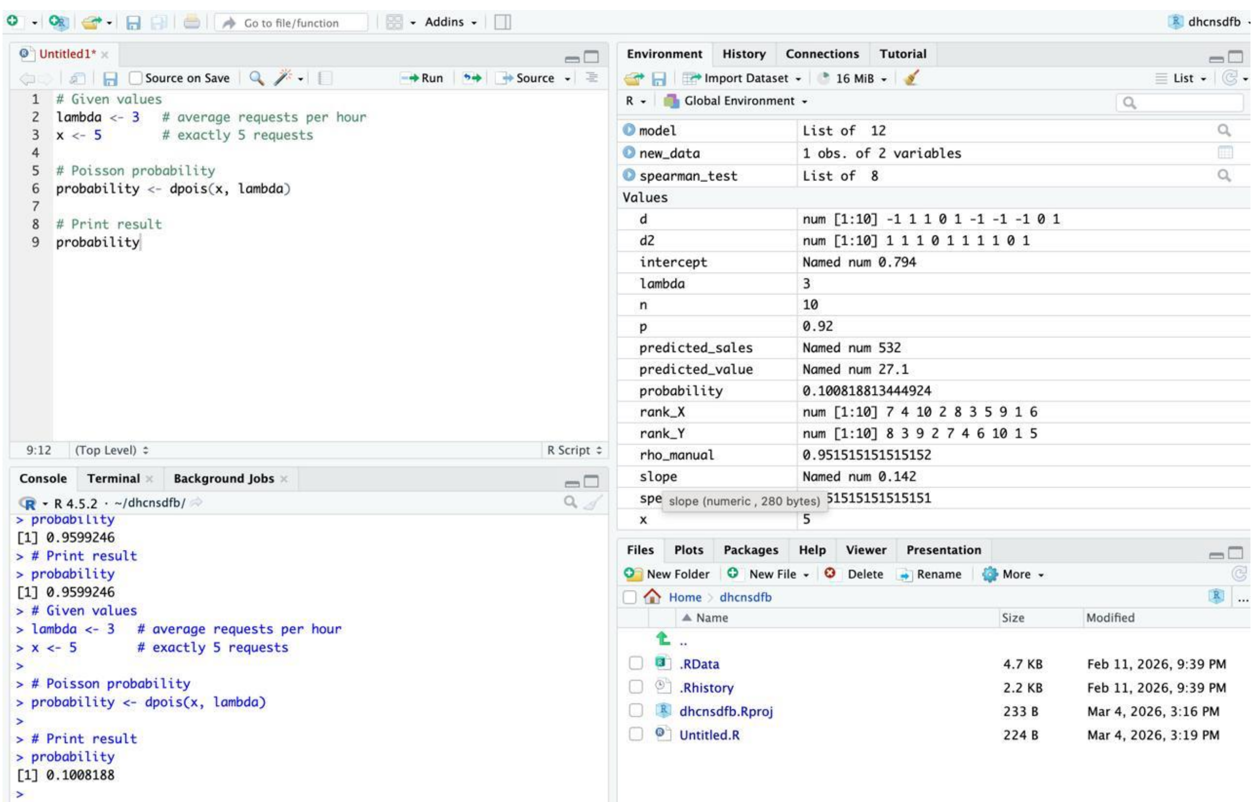
Print result

probability

Que-3:-

1. A help desk receives an average of 3 service requests every hour. Solve using Poisson distribution to find the probability that exactly 5 requests are received in a particular hour.

CODE:-



Que-3(code part):-

Given values

lambda <- 3 # average requests per hour x <- 5 # exactly 5 requests

Poisson probability

probability <- dpois(x, lambda)

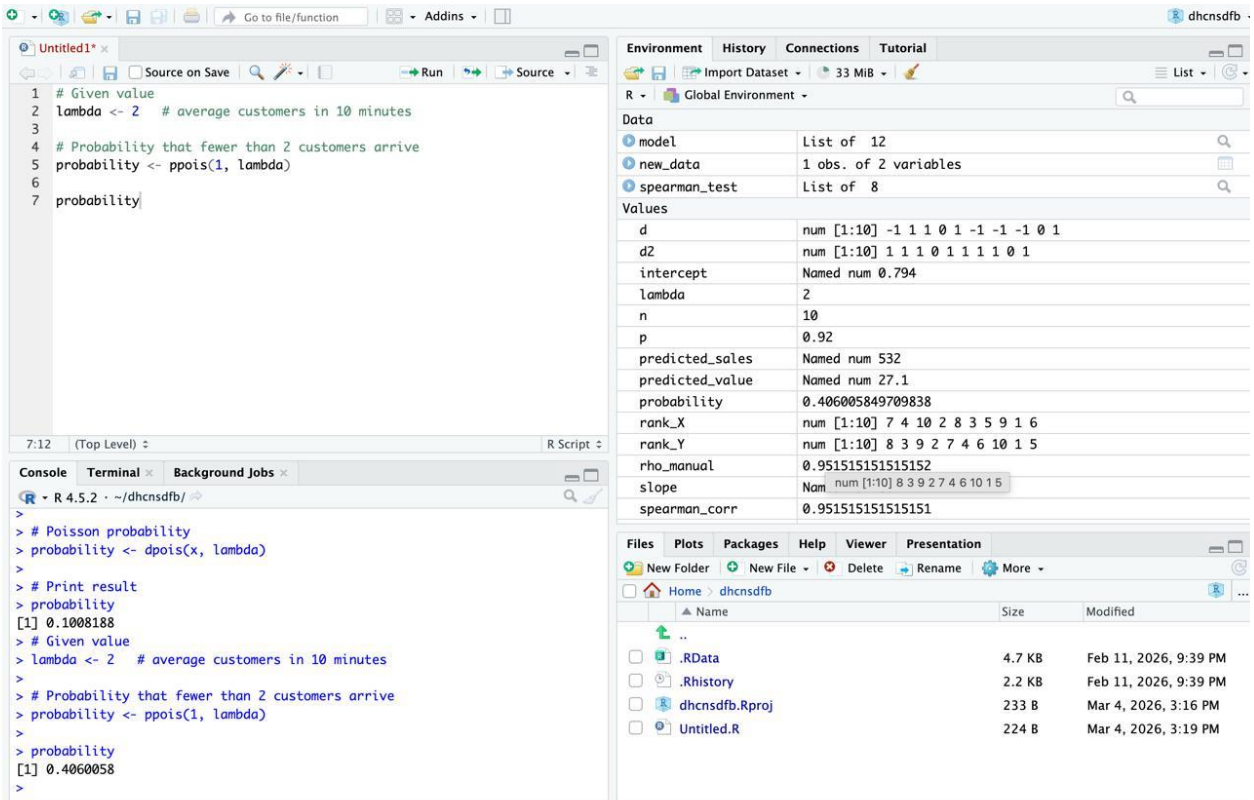
Print result

probability

Que-4:-

1. A railway reservation counter observes that an average of 2 customers arrive every 10 minutes. Solve using Poisson distribution to determine the probability that fewer than 2 customers arrive during a 10-minute interval.

Code:-



Question-4(code):-

Given value

lambda <- 2 # average customers in 10 minutes

Probability that fewer than 2 customers arrive

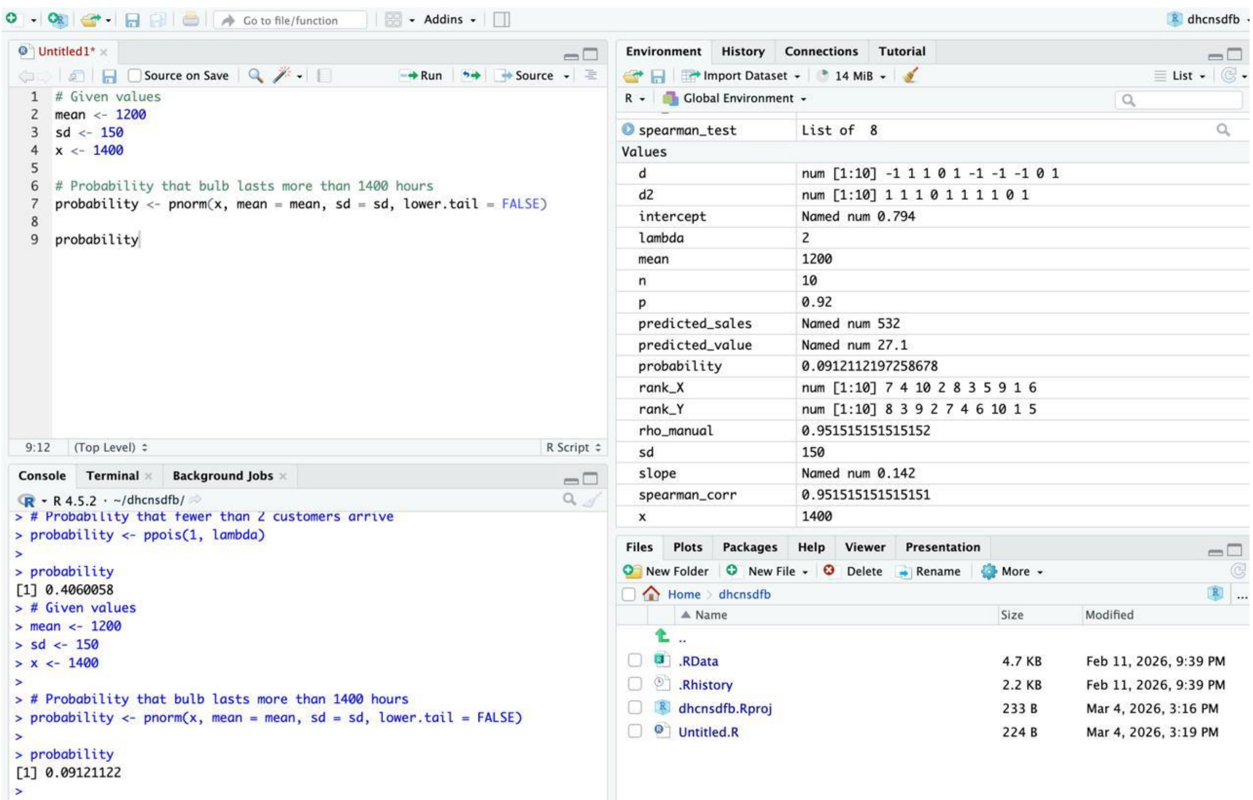
probability <- ppois(1, lambda)

Probability

Que-5:-

1. The lifetime of LED bulbs produced in a factory follows a normal distribution with mean 1200 hours and standard deviation 150 hours. Solve using Normal distribution to find the probability that a bulb lasts more than 1400 hours.

Code:-



Code-part:-

Given values

mean <- 1200 sd <- 150 x <- 1400

Probability that bulb lasts more than 1400 hours

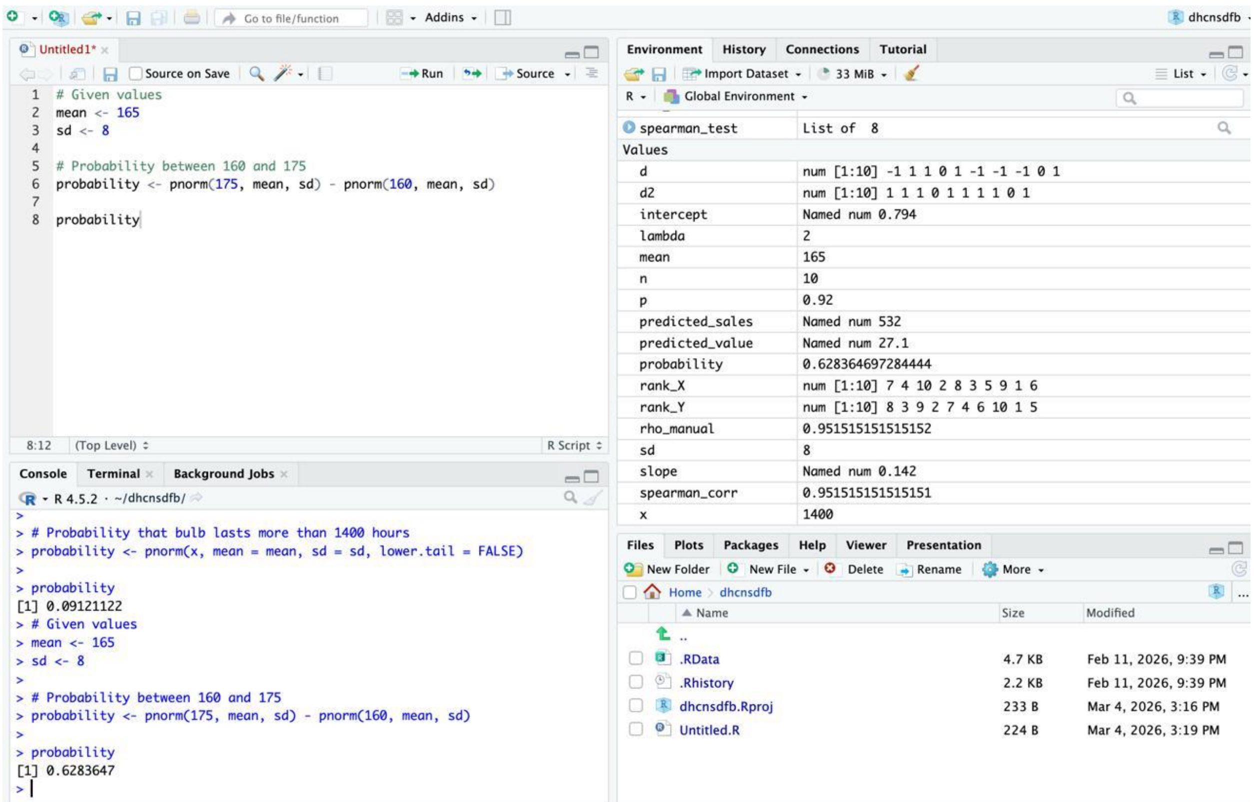
probability <- pnorm(x, mean = mean, sd = sd, lower.tail = FALSE)

Probability

Que-6:-

1. The heights of students in a college are normally distributed with mean 165 cm and standard deviation 8 cm. Solve using Normal distribution to determine the probability that a randomly selected student has height between 160 cm and 175 cm.

Code:-



Code-part:-

Given values

mean <- 165 sd <- 8

Probability between 160 and 175

probability <- pnorm(175, mean, sd) - pnorm(160, mean, sd)

probability